Iterative Constrained WLS Localization with Hybrid 1-D SA and TDOA Measurements

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ABSTRACT

Localization using hybrid one-dimensional space angle (SA) and time-difference-of-arrival (TDOA) measurements has broad potential in diverse applications. However, existing closed-form algorithms suffer from significant performance degradation under high noise levels because they inherently couple the noise from both measurement types. To overcome this limitation, we propose a novel iterative constrained weighted least-squares (ICWLS) algorithm. The method independently linearizes the SA and TDOA measurement equations before integrating them within a constrained weighted least-squares framework. This separation avoids cross-contamination of noise terms and directly addresses the primary weakness of prior works. Simulation results show that the proposed ICWLS algorithm consistently outperforms existing methods, achieving substantially higher localization accuracy, particularly in noisy environments.

***Index Terms*—** Space angle (SA), time-difference-of-arrival (TDOA), iterative constrained weighted least squares (ICWLS)

1. INTRODUCTION

Source localization is a core enabling technology for a broad range of applications, including intelligent transportation systems, emergency response, and defense operations. Among the various measurement modalities, time-difference-of-arrival (TDOA) and angle-of-arrival (AOA) are extensively employed [1, 2, 3, 4, 5]. Recently, localization based on one-dimensional (1-D) space angle (SA) measurements has received considerable attention due to its implementation advantages [6, 7, 8]. Unlike traditional AOA systems that require planar arrays to achieve three-dimensional localization, SA can be realized using a simple linear array, thereby reducing hardware complexity and resource consumption. Moreover, the integration of SA with TDOA measurements yields a hybrid localization framework that jointly exploits spatial and temporal information, offering improved robustness and localization accuracy.

Several closed-form algorithms have been proposed for this hybrid localization problem. Hu *et al.* [9] linearized the nonlinear SA equation using TDOA measurements, but this approach inevitably couples TDOA and SA noise. Xing *et al.* [10] improved WLS through a two-stage scheme that exploits quadratic constraint, yet the noise-coupling issue remains unresolved. To mitigate bias, Xing *et al.* later introduced the bias-subtracted deviation compensation (BSDC) and bias-reduced weighted total least-squares (BRWTLS) methods [11] to remove the bias inherent in the weighted total least-squares (WTLS) solutions.

A common characteristic of these methods is the substitution of TDOA measurements into the nonlinear SA equation to obtain a pseudolinear form. While this algebraic manipulation enables a closed-form solution, it introduces a fundamental drawback: the noise from TDOA and SA measurements becomes intrinsically coupled in the resulting equations.

This cross-contamination of noise terms significantly degrades estimator optimality. Performance deteriorates drastically in the presence of moderate or high measurement noise, as error propagation is exacerbated by the coupled noise structure [12]. This critical limitation motivates the development of a new localization algorithm that exploits independently the statistical properties of each measurement type. By decoupling SA and TDOA, error cross-contamination is mitigated, yielding more robust and accurate position estimates, particularly under adverse noise conditions.

To this end, we propose an iterative constrained weighted least-squares (ICWLS) algorithm. The key idea is to treat SA and TDOA equations as independent sources of information. The nonlinear SA equation is linearized from an iterative perspective [7] without introducing TDOA terms, while the TDOA equation is linearized using an existing approach [13]. The two resulting sets of linearized equations are then incorporated to form a constrained weighted least-squares (CWLS) problem. The final position estimate is obtained via a Lagrange multiplier method, which efficiently incorporates the geometric constraint relating the unknown source position to its range from a reference sensor. The overall solution is iterative, but simulations demonstrate that ICWLS converges rapidly (typically within a couple of iterations) and outperforms existing algorithms [9, 10, 11].

The main contributions of this work are as follows: we propose a decoupled formulation that eliminates noise coupling between SA and TDOA measurements, develop an ICWLS algorithm that significantly improves estimation accuracy under high-noise conditions, and demonstrate through simulations that the proposed method achieves both superior accuracy and rapid convergence compared with existing approaches.

Notation: The following notations are used throughout the paper. Bold uppercase and bold lowercase letters denote matrices and vectors, respectively. $\mathbf{I}_m$ is the $m \times m$ identity matrix and $\mathbf{0}_{m,n}$ is an $m \times n$ zero matrix. $\mathbb{E}(\cdot)$ denotes expectation and $\|\cdot\|$ is the ℓ_2 norm. $\mathbf{A}_{[i,:]}$ denotes the ith row of $\mathbf{A}$, and $\mathbf{A}(i,j)$ denotes the (i,j)th element of $\mathbf{A}$.

2. PROBLEM STATEMENT

Consider a localization scenario where a target at an unknown position $\mathbf{u} = [x, y, z]^T$ is observed by M linear arrays. The center of the i-th array is located at a known position $\mathbf{s}_i = [x_i, y_i, z_i]^T$ for

$i=1,\cdots,M$. Each array has a known orientation, described by a normalized attitude vector $\mathbf{g}_i \in \mathbb{R}^3$ [6],

$$\mathbf{g}_i = [\cos\theta_i\cos\phi_i, \sin\theta_i\cos\phi_i, \sin\phi_i]^T, \tag{1}$$

where θ_i and ϕ_i are the azimuth and elevation angles of the attitude vector, respectively. The system acquires two types of measurements: SA and TDOA. The SA measurement from the i-th array, denoted by ψ_i, is modeled as

$$\psi_i = \psi_i^0 + e_i, \quad i=1,\cdots,M, \tag{2}$$

where ψ_i^0 is the true SA, given by

$$\psi_i^0 = \cos^{-1}\left(\frac{\mathbf{g}_i^T(\mathbf{u}-\mathbf{s}_i)}{\|\mathbf{u}-\mathbf{s}_i\|}\right), \tag{3}$$

and e_i is the measurement noise. The noise vector $\mathbf{e}=[e_1,\cdots,e_M]^T$ is assumed to be zero-mean Gaussian distributed with covariance matrix $\mathbf{Q}_e$. Since TDOA is interchangeable to range-difference-of-arrival (RDOA) given the known speed of light, in the following, TDOA and RD are equivalent. Taking the first array as the reference, the RDOA measurement between the i-th array and the reference array is

$$r_{i1} = \|\mathbf{u}-\mathbf{s}_i\| - \|\mathbf{u}-\mathbf{s}_1\| + n_{i1}, \quad i=2,\cdots,M, \tag{4}$$

where n_{i1} is the measurement noise. The noise vector $\mathbf{n} = [n_{21},\cdots,n_{M1}]^T$ is modeled as a zero-mean Gaussian random vector with covariance matrix $\mathbf{Q}_d$. The goal is to estimate the unknown target position $\mathbf{u}$ from the set of noisy measurements $\{\psi_i\}_{i=1}^M$ and $\{r_{i1}\}_{i=2}^M$.

3. EXISTING WORK

The SA equation is highly nonlinear and therefore not solvable directly. In [9, 10, 11], the TDOA measurement is used to linearize the SA equation, leading to coupled TDOA and SA noise in their derivations. Here, we present the derivation of WLS from [9] as an example.

First, the TDOA equation is expressed as

$$r_{i1} + \|\mathbf{u}-\mathbf{s}_1\| = \|\mathbf{u}-\mathbf{s}_i\| + n_{i1}. \tag{5}$$

Squaring both sides and ignoring the second-order noise terms yield:

$$\begin{aligned} r_{i1}^2 + 2r_{i1}\|\mathbf{u}-\mathbf{s}_1\| \approx 2\mathbf{u}^T(\mathbf{s}_1-\mathbf{s}_i) + \|\mathbf{s}_i\|^2 - \|\mathbf{s}_1\|^2 + \\ 2\|\mathbf{u}-\mathbf{s}_i\|n_{i1}, \end{aligned} \tag{6}$$

which is equivalent to

$$\begin{aligned} \mathbf{u}^T(\mathbf{s}_i-\mathbf{s}_1) + r_{i1}\|\mathbf{u}-\mathbf{s}_1\| \approx 0.5(\|\mathbf{s}_i\|^2 - \|\mathbf{s}_1\|^2 - r_{i1}^2) + \\ \|\mathbf{u}-\mathbf{s}_i\|n_{i1}. \end{aligned} \tag{7}$$

Second, taking the cos operation for both sides of (3) yields

$$\cos\psi_i^0\|\mathbf{u}-\mathbf{s}_i\| = \mathbf{g}_i^T(\mathbf{u}-\mathbf{s}_i). \tag{8}$$

Substituting the following TDOA equation

$$\|\mathbf{u}-\mathbf{s}_i\| = r_{i1} + \|\mathbf{u}-\mathbf{s}_1\| - n_{i1}, \tag{9}$$

into (8) yields

$$\cos\psi_i^0(r_{i1} + \|\mathbf{u}-\mathbf{s}_1\| - n_{i1}) = \mathbf{g}_i^T(\mathbf{u}-\mathbf{s}_i). \tag{10}$$

When the SA measurement noise is taken into consideration and under small-noise conditions, we have

$$\cos\psi_i^0 = \cos(\psi_i - e_i) \approx \cos\psi_i + \sin\psi_i e_i. \tag{11}$$

Substituting (11) into (10) yields

$$\begin{aligned} \mathbf{g}_i^T\mathbf{u} - \cos\psi_i\|\mathbf{u}-\mathbf{s}_1\| \approx \mathbf{g}_i^T\mathbf{s}_i + r_{i1}\cos\psi_i + \\ \sin\psi_i\|\mathbf{u}-\mathbf{s}_i\|e_i - \cos\psi_i n_{i1}. \end{aligned} \tag{12}$$

Finally, stacking equations (7) and (12) yields

$$\mathbf{G}\mathbf{x} \approx \mathbf{h} + \mathbf{B}\boldsymbol{\epsilon} \tag{13}$$

where

$$\mathbf{G}_{[i-1,:]} = [(\mathbf{s}_i-\mathbf{s}_1)^T, r_{i1}], i=2,\cdots,M, \tag{14a}$$
$$\mathbf{h}(i-1) = 0.5(\|\mathbf{s}_i\|^2 - \|\mathbf{s}_1\|^2 - r_{i1}^2), i=2,\cdots,M, \tag{14b}$$
$$\mathbf{G}_{[M-1+i,:]} = [\mathbf{g}_i^T, -\cos\psi_i], i=1,\cdots,M, \tag{14c}$$
$$\mathbf{h}(M-1+i) = \mathbf{g}_i^T\mathbf{s}_i + \cos\psi_i r_{i1}, i=1,\cdots,M, \tag{14d}$$
$$\mathbf{x} = [\mathbf{u}^T, \|\mathbf{u}-\mathbf{s}_1\|]^T, \boldsymbol{\epsilon} = [\mathbf{n}^T, \mathbf{e}^T]^T, \tag{14e}$$
$$\mathbf{B} = \begin{bmatrix} \mathbf{B}_{11} & \mathbf{0} \\ \mathbf{B}_{21} & \mathbf{B}_{22} \end{bmatrix} \in \mathbb{R}^{(2M-1)\times(2M-1)}, \tag{14f}$$
$$\mathbf{B}_{11} = \mathrm{diag}([\|\mathbf{u}-\mathbf{s}_2\|, \ldots, \|\mathbf{u}-\mathbf{s}_M\|]), \tag{14g}$$
$$\mathbf{B}_{21} = \begin{bmatrix} 0 & \cdots & 0 \\ -\cos\psi_2 & \cdots & 0 \\ 0 & \ddots & 0 \\ 0 & \cdots & -\cos\psi_M \end{bmatrix} \in \mathbb{R}^{M\times(M-1)}, \tag{14h}$$
$$\mathbf{B}_{22} = \mathrm{diag}([\|\mathbf{u}-\mathbf{s}_1\|\sin\psi_1, \cdots, \|\mathbf{u}-\mathbf{s}_M\|\sin\psi_M]). \tag{14i}$$

Note that r_{i1} is not defined for $i=1$ in (14d). In order to obtain consistent expression, we defined $r_{11}=0$.

Remark 1 *From (12), it is obvious that both the SA noise and the TDOA noise coexist in the linearized SA equation, which leads to performance degradation when noise variances are not sufficiently small [14].*

4. PROPOSED ALGORITHM

The core idea of the proposed ICWLS is to linearize the SA and TDOA measurement equations separately to avoid the noise cross-contamination problem discussed previously. These linearized equations are then integrated into a CWLS framework to obtain the position estimate.

The SA formulation starts from the noise-free equation in (8) and the first-order approximation of the cosine term

$$\cos\psi_i^0\|\mathbf{u}-\mathbf{s}_i\| = \mathbf{g}_i^T(\mathbf{u}-\mathbf{s}_i), \tag{15}$$
$$\cos\psi_i^0 = \cos(\psi_i - e_i) \approx \cos\psi_i + \sin\psi_i e_i. \tag{16}$$

Substituting (16) into (15) yields

$$\cos\psi_i\|\mathbf{u}-\mathbf{s}_i\| + \sin\psi_i\|\mathbf{u}-\mathbf{s}_i\|e_i \approx \mathbf{g}_i^T(\mathbf{u}-\mathbf{s}_i). \tag{17}$$

To linearize this equation with respect to the unknown position $\mathbf{u}$, the norm term $\|\mathbf{u}-\mathbf{s}_i\|$ is expressed as $\frac{(\mathbf{u}-\mathbf{s}_i)^T(\mathbf{u}-\mathbf{s}_i)}{\|\mathbf{u}-\mathbf{s}_i\|}$. This substitution leads to

$$(\mathbf{u}-\mathbf{s}_i)^T\left(\mathbf{g}_i - \cos\psi_i\boldsymbol{\rho}_{\mathbf{u}-\mathbf{s}_i}\right) \approx \sin\psi_i\|\mathbf{u}-\mathbf{s}_i\|e_i, \tag{18}$$

where $\boldsymbol{\rho}_{\mathbf{u}-\mathbf{s}_i} = \frac{\mathbf{u}-\mathbf{s}_i}{\|\mathbf{u}-\mathbf{s}_i\|}$. For consistency with the subsequent TDOA formulation, equation (18) is rearranged by introducing the reference sensor position $\mathbf{s}_1$,

$$\begin{aligned}&(\mathbf{u}-\mathbf{s}_1)^T\left(\mathbf{g}_i - \cos\psi_i \boldsymbol{\rho}_{\mathbf{u}-\mathbf{s}_i}\right)\\ &\approx (\mathbf{s}_i-\mathbf{s}_1)^T\left(\mathbf{g}_i - \cos\psi_i \boldsymbol{\rho}_{\mathbf{u}-\mathbf{s}_i}\right) + \sin\psi_i\|\mathbf{u}-\mathbf{s}_i\|e_i. \qquad (19)\end{aligned}$$

The TDOA equation is rewritten as

$$r_{i1} + \|\mathbf{u}-\mathbf{s}_1\| = \|\mathbf{u}-\mathbf{s}_1+\mathbf{s}_1-\mathbf{s}_i\| + n_{i1}, \qquad (20)$$

which can be linearized using a standard algebraic manipulation [13]. Squaring both sides of (20) and neglecting the second-order noise term give

$$\begin{aligned}r_{i1}^2 + 2r_{i1}\|\mathbf{u}-\mathbf{s}_1\| &\approx 2(\mathbf{u}-\mathbf{s}_1)^T(\mathbf{s}_1-\mathbf{s}_i)\\ &\quad + \|\mathbf{s}_1-\mathbf{s}_i\|^2 + 2\|\mathbf{u}-\mathbf{s}_i\|n_{i1}. \qquad (21)\end{aligned}$$

Rearranging the terms yields the following equation,

$$\begin{aligned}&(\mathbf{u}-\mathbf{s}_1)^T(\mathbf{s}_i-\mathbf{s}_1) + r_{i1}\|\mathbf{u}-\mathbf{s}_1\|\\ &\approx 0.5(\|\mathbf{s}_1-\mathbf{s}_i\|^2 - r_{i1}^2) + \|\mathbf{u}-\mathbf{s}_i\|n_{i1}. \qquad (22)\end{aligned}$$

By stacking the linearized SA equations from (19) for $i = 1, \cdots, M$ and the linearized TDOA equations from (22) for $i = 2, \cdots, M$, a combined linear system is formed,

$$\mathbf{A}\mathbf{y} \approx \mathbf{b} + \mathbf{D}\boldsymbol{\epsilon} \qquad (23)$$

where the auxiliary vector $\mathbf{y}$ is $\mathbf{y} = [(\mathbf{u}-\mathbf{s}_1)^T, \|\mathbf{u}-\mathbf{s}_1\|]^T$, and noise vector $\boldsymbol{\epsilon}$ is same as defined in (14e). The matrices $\mathbf{A}$ and $\mathbf{D}$, and vector $\mathbf{b}$ are defined as follows.

$$\mathbf{A}_{[i-1,:]} = [(\mathbf{s}_i-\mathbf{s}_1)^T, r_{i1}], i = 2, \cdots, M, \qquad (24a)$$

$$\mathbf{b}(i-1) = 0.5(\|\mathbf{s}_1-\mathbf{s}_i\|^2 - r_{i1}^2), i = 2, \cdots, M, \qquad (24b)$$

$$\mathbf{A}_{[M-1+i,:]} = [(\mathbf{g}_i - \cos\psi_i\boldsymbol{\rho}_{\mathbf{u}-\mathbf{s}_i})^T, 0], i = 1, \cdots, M, \qquad (24c)$$

$$\mathbf{b}(M-1+i) = (\mathbf{s}_i-\mathbf{s}_1)^T(\mathbf{g}_i - \cos\psi_i\boldsymbol{\rho}_{\mathbf{u}-\mathbf{s}_i}), i = 1, \cdots, M, \qquad (24d)$$

$$\mathbf{D} = \text{blkdiag}(\mathbf{D}_1, \mathbf{D}_2), \qquad (24e)$$

$$\mathbf{D}_1 = \text{diag}([\|\mathbf{u}-\mathbf{s}_2\|, \cdots, \|\mathbf{u}-\mathbf{s}_M\|]), \qquad (24f)$$

$$\mathbf{D}_2 = \text{diag}([\|\mathbf{u}-\mathbf{s}_1\|\sin\psi_1, \cdots, \|\mathbf{u}-\mathbf{s}_M\|\sin\psi_M]). \qquad (24g)$$

This system leads to a CWLS problem for estimating $\mathbf{y}$,

$$\min_{\mathbf{y}} \quad (\mathbf{A}\mathbf{y}-\mathbf{b})^T\mathbf{W}(\mathbf{A}\mathbf{y}-\mathbf{b}) \qquad (25a)$$

$$\text{s.t.} \quad \mathbf{y}^T\mathbf{P}\mathbf{y} = 0 \qquad (25b)$$

The weight matrix $\mathbf{W}$ is the inverse of the noise covariance matrix, i.e.,

$$\begin{aligned}\mathbf{W} &= (\mathbb{E}[\mathbf{D}\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T\mathbf{D}^T])^{-1}\\ &= [\text{blkdiag}(\mathbf{D}_1\mathbf{Q}_d\mathbf{D}_1, \mathbf{D}_2\mathbf{Q}_e\mathbf{D}_2)]^{-1}, \qquad (26)\end{aligned}$$

where $\mathbf{blkdiag}(\cdot)$ denotes the block diagonal matrix. The constraint in (25b) with $\mathbf{P} = \text{diag}([1,1,1,-1]^T)$ enforces the geometric relationship between the first three elements of $\mathbf{y}$ and its fourth element.

The CWLS problem in (25) can be solved using the method of Lagrange multipliers. The Lagrangian is given by

$$\mathcal{L}(\mathbf{y},\lambda) = (\mathbf{A}\mathbf{y}-\mathbf{b})^T\mathbf{W}(\mathbf{A}\mathbf{y}-\mathbf{b}) + \lambda\mathbf{y}^T\mathbf{P}\mathbf{y}. \qquad (27)$$

Setting the gradient of $\mathcal{L}(\mathbf{y},\lambda)$ with respect to $\mathbf{y}$ to zero yields the solution

$$\mathbf{y} = (\mathbf{A}^T\mathbf{W}\mathbf{A} + \lambda\mathbf{P})^{-1}\mathbf{A}^T\mathbf{W}\mathbf{b}, \qquad (28)$$

where the Lagrange multiplier λ is determined by substituting (28) back into the constraint (25b) and solving the resulting equation, as detailed in [14].

The matrices $\mathbf{A}$ and $\mathbf{W}$, and vector $\mathbf{b}$ depend on the unknown target position $\mathbf{u}$, impeding the settlement of the CWLS problem. Therefore, an iterative procedure is employed. Given an initial estimate of the target position, $\mathbf{u}^{(0)}$, the matrices and vector are computed, and the CWLS problem is solved to obtain an updated estimate by

$$\begin{aligned}\mathbf{y}^{(k)} = (&\mathbf{A}^{(k-1)T}\mathbf{W}^{(k-1)}\mathbf{A}^{(k-1)} + \lambda\mathbf{P})^{-1}\\ &\cdot \mathbf{A}^{(k-1)T}\mathbf{W}^{(k-1)}\mathbf{b}^{(k-1)}. \qquad (29)\end{aligned}$$

This process is repeated until convergence. The WLS solution from [9] serves as a robust initial value. The iteration terminates when the change in the position estimate is below a predefined threshold τ, or a maximum number of iterations $K_{\max}$ is reached:

$$\|\mathbf{u}^{(k)} - \mathbf{u}^{(k-1)}\| \leq \tau, \quad \text{or} \quad k \geq K_{\max}. \qquad (30)$$

Algorithm 1: ICWLS Algorithm for Hybrid 1-D SA and TDOA Localization

Data: $r_{i1}, \psi_i, \mathbf{s}_i, K_{max}, \tau$;
Result: $\mathbf{u}_{\text{out}}$;
$\mathbf{u}^0 \leftarrow$ calculate the WLS algorithm in [9];
$k \leftarrow 1$;
while $k \leq K_{max}$ **do**
  $\mathbf{A}^{k-1}, \mathbf{b}^{k-1} \leftarrow$ calculate (24);
  $\mathbf{W}^{k-1} \leftarrow$ calculate (26);
  $\mathbf{u}^k \leftarrow$ calculate (29);
  if *(30) is satisfied* **then**
    break;
  end
  $k \leftarrow k + 1$;
end

5. SIMULATION RESULTS

This section evaluates the performance of the proposed ICWLS algorithm through numerical simulations. The algorithm is benchmarked against several existing methods: WLS [9], TSWLS [10], BSDC [11], and BRWTLS [11]. The Cramér-Rao Lower Bound (CRLB) is also included as a theoretical performance benchmark [10]. Performance is measured in terms of the root-mean-square error (RMSE), averaged over 1000 Monte Carlo trials for each of 10 randomly generated target positions.

The target position $\mathbf{u}$ is randomly drawn from the volume $[-2000, 2000] \times [-2000, 2000] \times [0, 1000]$ m. The array parameters are detailed in Table 1. The SA measurement noise vector $\mathbf{e}$ is modeled as zero-mean Gaussian with covariance $\mathbf{Q}_e = \sigma_a^2\mathbf{I}_M$. The TDOA noise vector $\mathbf{n}$ is also zero-mean Gaussian with covariance $\mathbf{Q}_d = \sigma_d^2\mathbf{R}$, where $\mathbf{R}$ is a symmetric matrix with diagonal elements equal to 1 and off-diagonal elements equal to 0.5, reflecting correlated noise among TDOA measurements [13]. For the proposed ICWLS algorithm, the maximum number of iterations is

Table 1. Parameters of Arrays for Hybrid Localization

No.	1	2	3	4
x (m)	500	500	-500	-500
y (m)	500	-500	500	-500
z (m)	50	100	150	200
θ (rad)	-2.61	2.78	-0.54	-0.17
ϕ (rad)	0.92	0.19	-0.64	0.16

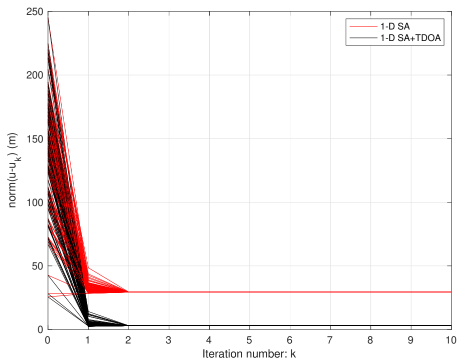

Fig. 1. Estimate bias versus iteration number (fixed $\sigma_a = -20$ dB rad and $\sigma_d = 0$ dB m).

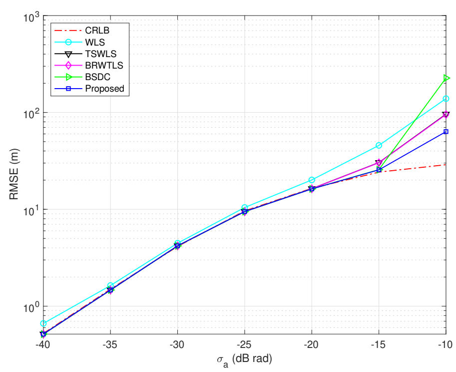

Fig. 2. RMSE versus σ_a (fixed $\sigma_d = 0$ dB m).

set to $K_{\max} = 2$, and the convergence threshold is $\tau = 0.01$ m. Two logarithmic units are defined as: dB m $= 10 * \log_{10}(\mathrm{m})$, and dB rad $= 10 * \log_{10}(\mathrm{rad})$.

Fig. 1 illustrates the convergence behavior of the proposed ICWLS algorithm. The performance of an iterative WLS (IWLS) algorithm that uses only SA measurements [7] is included for comparison. The simulation is conducted for a fixed target at $\mathbf{u} = [624.8, 719, 975]^T$ m, with 100 different initial positions generated by adding a random vector (with entries bounded by ± 150 m) to the true position. The results demonstrate two key points: First, the proposed ICWLS algorithm exhibits rapid convergence, typically reaching a stable solution within two iterations. This fast convergence makes it computationally efficient for practical applications. Second, the integration of TDOA measurements significantly enhances localization accuracy, as evidenced by the substantially lower bias of ICWLS compared to the SA-only IWLS.

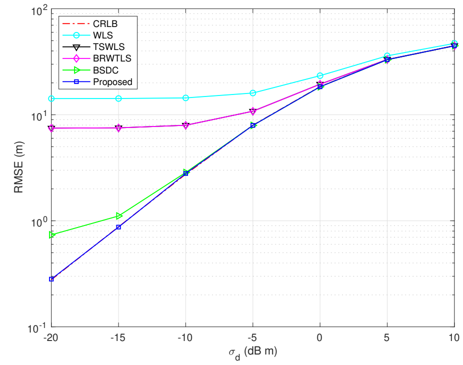

Fig. 3. RMSE versus σ_d (fixed $\sigma_a = -20$ dB rad).

Fig. 2 presents the RMSE performance of all algorithms as a function of the SA noise standard deviation σ_a, with the TDOA noise fixed at $\sigma_d = 0$ dB m. The WLS algorithm is clearly suboptimal, as its RMSE lies above the CRLB even at low noise levels. While TSWLS, BSDC, and BRWTLS show improved performance, the proposed ICWLS algorithm achieves the lowest RMSE, particularly for $\sigma_a > -15$ dB rad (approximately $1.81°$). This demonstrates the robustness of our method against increasing SA noise.

Fig. 3 shows the RMSE performance versus the TDOA noise standard deviation σ_d, with a fixed SA noise of $\sigma_a = -20$ dB rad (approximately $0.57°$). The proposed ICWLS algorithm consistently outperforms all other methods and its performance curve closely follows the CRLB. A critical observation is that the performance of the competing algorithms, WLS, TSWLS, BSDC and BRWTLS, saturates and fails to reach the CRLB even at very low TDOA noise levels (e.g., $\sigma_d < -10$ dB m). This behavior confirms our findings: because these methods mix SA and TDOA noise in their formulation, their accuracy becomes limited by the fixed, dominant SA noise ($\sigma_a = -20$ dB rad). In contrast, the proposed ICWLS algorithm, by decoupling the noise sources, effectively utilizes the high-quality TDOA information and is not hampered by this performance bottleneck.

6. CONCLUSIONS

This paper addressed the problem of source localization using hybrid SA and TDOA measurements. We proposed a novel ICWLS algorithm that fundamentally avoids the noise coupling issue inherent in state-of-the-art methods by processing the two measurement types separately. Extensive simulations validated the theoretical advantages of this approach, demonstrating that the proposed ICWLS algorithm consistently outperforms existing methods, especially under significant noise conditions. These results underscore the value of preventing noise cross-contamination for developing robust hybrid localization systems.

7. REFERENCES

[1] Yanbin Zou and Huaping Liu, "A simple and efficient iterative method for TOA localization," in *Proc. IEEE Int. Conf. Acoust. Speech, Signal Process.*, 2020, pp. 4881–4884.

[2] Yanbin Zou, Yangpeng Xiao, Zekai Zhang, and Huaping Liu, "TDOA localization via fixed-point iteration," in *IEEE Inter-*

national Conference on Acoustics, Speech and Signal Processing (ICASSP), 2025, pp. 1–5.

[3] Yue Wang and KC Ho, "An asymptotically efficient estimator in closed-form for 3-D AOA localization using a sensor network," *IEEE Trans. Wireless Commun.*, vol. 14, no. 12, pp. 6524–6535, 2015.

[4] Yanbin Zou, Liehu Wu, Jingna Fan, and Huaping Liu, "A convergent iteration method for 3-D AOA localization," *IEEE Trans. Veh. Technol.*, vol. 72, no. 6, pp. 8267–8271, Jun. 2023.

[5] Jihao Yin, Qun Wan, Shiwen Yang, and KC Ho, "A simple and accurate TDOA-AOA localization method using two stations," *IEEE Signal Process. Lett.*, vol. 23, no. 1, pp. 144–148, 2016.

[6] Yimao Sun, K. C. Ho, Lin Gao, Jifeng Zou, Yanbing Yang, and Liangyin Chen, "Three dimensional source localization using arrival angles from linear arrays: Analytical investigation and optimal solution," *IEEE Trans. Signal Process.*, vol. 70, pp. 1864–1879, 2022.

[7] Yanbin Zou, Yangpeng Xiao, Wenbo Wu, Yimao Sun, and Huaping Liu, "A new iterative weighted least squares algorithm for 1-D SA localization," *IEEE Signal Process. Lett.*, vol. 32, pp. 1740–1744, 2025.

[8] Yonghua Chen, Hua Yu, Jie Li, Qisen Wang, Fei Ji, and Fangjiong Chen, "Three-dimensional source localization based on 1-D AOA measurements: Low-complexity and effective estimator," *IEEE Transactions on Instrumentation and Measurement*, vol. 72, pp. 1–15, 2023.

[9] Qiong Hu, Yixun Peng, Qun Wan, Zepeng Hu, Zongquan Wang, and Yujun Zhu, "A novel 3-D localization scheme using 1-D AOA and TDOA measurements," in *IEEE 94th Vehicular Technology Conference (VTC2021-Fall)*, 2021, pp. 1–5.

[10] Tianyi Xing, Yimao Sun, Lihua Ni, Xiangyu Peng, Kehao Zhang, and Qun Wan, "Solution and analysis for 3-D localization in closed-form integrating SA and TDOA measurements," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024, pp. 8831–8835.

[11] Tianyi Xing, Yimao Sun, Kehao Zhang, Xiangyu Peng, Lihua Ni, Ziqiang Wang, Ning Liu, and Qun Wan, "3-D localization integrating 1-D SA and TDOA measurements: Analysis and bias-reduced solutions," *IEEE Trans. Veh. Technol.*, 2025.

[12] Yanbin Zou, Xiaofei Li, Shiru Chen, Yihan Wang, and Yimao Sun, "A noise-decoupled wls solution for hybrid AOA-TDOA localization in the presence of sensor position errors," *Signal Processing*, vol. 243, pp. 110481, 2026.

[13] Yiu Tong Chan and K. C. Ho, "A simple and efficient estimator for hyperbolic location," *IEEE Transa. Signal Process.*, vol. 42, no. 8, pp. 1905–1915, 1994.

[14] Yanbin Zou, Wenbo Wu, Jingna Fan, and Huaping Liu, "Constrained weighted least-squares algorithms for 3-D AOA-based hybrid localization," *IEEE Open Journal of Signal Processing*, vol. 5, pp. 436–448, 2024.